

1. Administrative & Study Identification

Full Study Title: Study of Alpelisib (BYL719) in Combination With Trastuzumab and Pertuzumab as Maintenance Therapy in Patients With HER2-positive Advanced Breast Cancer With a PIK3CA Mutation
Short Title/Acronym: EPIK-B2 **Protocol Number:** CBYL719G12301 **NCT Number:** NCT04208178 **Posting ID:** 1096663396468018684 **Sponsor/Company Name:** Novartis Pharmaceuticals (Novartis) **Investigational Product:** Alpelisib (BYL719) / Trastuzumab / Pertuzumab (*Note: Trastuzumab ATC Code is L01FD01; Trade names include Herceptin, Enhertu, Ontruzant, Trazimera, Herzuma, Phesgo, Kanjinti, Herceptin Hylecta, Ogivri, KADCYLA*) **Phase of Development:** Phase III **Medical Monitor:** Novartis Pharmaceuticals Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To evaluate the efficacy and safety of alpelisib compared to an alpelisib matching-placebo in combination with trastuzumab and pertuzumab as a maintenance treatment. * **Secondary Objectives:** To evaluate Progression-Free Survival (PFS) associations with PIK3CA mutation status measured via ctDNA at baseline, Overall Survival, and Health-Related Quality of Life (HRQOL) utilizing the FACT-B 37-item instrument.

2.2 Background & Rationale To address the clinical need for effective maintenance therapies in patients with HER2-positive advanced breast cancer whose tumor harbors a PIK3CA driver mutation immediately post-induction therapy (taxane + trastuzumab + pertuzumab). The study evaluates if alpelisib combined with trastuzumab and pertuzumab provides superior clinical benefit compared to placebo.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind (Part 2 uses matching placebo; Part 1 is an open-label safety run-in) **Randomization:** Randomized 1:1 (in Part 2)

3.2 Planned Enrollment & Duration **Trial Status:** Active, not recruiting (ACTIVE_NOT_RECRUITING) **Target \$N\$:** 20 participants. Recruitment was permanently halted as of 07-Dec-2022 due to the evolving treatment landscape and changing paradigms for HER-2 positive breast cancer therapy. **Number of Sites:** Multicenter trial (Global, including sites in the US, Europe, and Lebanon)

Estimated Study Start: August 2020 **Estimated Primary Completion:** Halted in December 2022

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participant has histologically-confirmed HER2-positive breast cancer that is advanced (locoregionally not amenable to surgery or is metastatic). 2. Participant has received pre-study induction therapy with up to and including a maximum of 8 cycles of a taxane (docetaxel, paclitaxel, or nab-paclitaxel), plus trastuzumab and pertuzumab. 3. Participant has an Eastern Cooperative Oncology Group (ECOG) performance status 0 or 1. 4. Participant has a PIK3CA mutation(s) present in tumor prior to enrollment.

4.2 Exclusion Criteria 1. Participant with inflammatory breast cancer at screening. 2. Participant with evidence of disease progression during the pre-study induction therapy and prior to the first dose of alpelisib. 3. Participant with an established diagnosis of diabetes mellitus type I or uncontrolled type II based on fasting plasma glucose (FPG) and HbA1c. 4. Participant has a known history of acute pancreatitis within 1 year of screening or past medical history of chronic pancreatitis.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Alpelisib (or placebo) combined with trastuzumab and pertuzumab, administered every 3 weeks per cycle. Part 1 evaluates multiple visits for dose escalation to confirm the recommended Phase 3 dose. **Route of Administration:** Oral (Alpelisib) and Intravenous/Subcutaneous (Trastuzumab, Pertuzumab). **Duration of Treatment:** Until disease progression or up to approximately 38 months (Part 2 randomized treatment duration target).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care measures per protocol. **Prohibited:** Cross-over from the placebo arm to the experimental arm was not permitted, though control patients could continue trastuzumab and pertuzumab if deriving clinical benefit.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | 2-4 Weeks | Informed Consent, Medical History, PIK3CA mutation confirmation | | **Baseline** | Day 0 | Randomization, First Dose, baseline ECG, tumor assessment | | **Interim** | Every 3 Weeks | Treatment cycles, Safety Labs, Efficacy Assessment. Electrocardiogram (ECG) and tumor imaging every 12

weeks | | **End of Study** | Variable | Final Vital Signs, exit follow-up for cancer status and survival upon progression |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-Free Survival (PFS) in Part 2; identification of Dose-Limiting Toxicities (DLTs) to confirm the RP3D in Part 1. **Measurement Method:** Assessment using RECIST 1.1 criteria.

7.2 Secondary & Exploratory Endpoints * Health-Related Quality of Life (HRQOL) assessed via the FACT-B 37-item instrument. * Overall Survival (OS). * Association between baseline PIK3CA mutation status in ctDNA and PFS.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Focus on common PI3K inhibitor toxicities including hyperglycemia, diarrhea, nausea, and rash. **Serious Adverse Events (SAEs):** Routine monitoring with ECG tracking every 12 weeks for cardiac safety. **Data Monitoring Committee (DMC):** Governed by Novartis and independent monitoring panels.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for efficacy, Safety Set for evaluating AE/SAEs. **Sample Size Justification:** Powered for PFS benefit in the PIK3CA mutated patient subset (though study was ultimately halted). **Handling of Missing Data:** Standard oncology trial censoring applied for survival (PFS/OS) analyses.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular central and onsite monitoring of trial endpoints and dosing safety. **Audit Trail:** Clinical data documented and anonymized via secure databases, managed by Novartis.

11. Study Identifiers & Governance

Primary ID: CBYL719G12301 **Registry Number:** NCT04208178 **Posting ID:** 1096663396468018684 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** EPIK-B2 **Official Regulatory Title:** A Two Part, Phase III, Multicenter, Randomized (1:1), Double-blind,

Placebo-controlled Study to Assess the Efficacy and Safety of Alpelisib (BYL719) in Combination With Trastuzumab and Pertuzumab as Maintenance Therapy in Patients With HER2-positive Advanced Breast Cancer With a PIK3CA Mutation

12. Clinical Framework (Breast Cancer Specific)

Primary Condition: Breast Cancer (Disease Area: Advanced HER2+ Breast Cancer) **Preferred Name:** Malignant neoplasm of breast (Preferred UMLS Name: HER2-positive, PIK3CA-mutated Advanced Breast Cancer) **Diagnosis Threshold:** PIK3CA mutation(s) present and confirmed in tumor **Medical Methodology:** Targeted Maintenance Therapy **Optimization Target:** Recommended phase 3 dose (RP3D) of Alpelisib

13. Study Procedures & Methodology

Management Pathway: Maintenance pathway following taxane + HER2-targeted induction **Education Protocol (HCP):** Protocol-specific safety monitoring and reporting (e.g., hyperglycemia) **Education Protocol (Patient):** Treatment adherence and adverse event management **Quality Audit Mechanism:** Site monitoring and patient treatment/drug adherence tracking

14. Observation & Follow-Up Schedule

Total Duration: Up to approximately 38 months **Onsite Visit Cadence:** Every 3 weeks corresponding with treatment cycles **Remote Monitoring Frequency:** Periodic safety follow-up **Checklist Review Interval:** Every 12 weeks for routine ECG and tumor imaging/assessment

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	____	[DD-MMM-YYYY]				